

AP Precalculus: 2026–2027

Unit 3: Trigonometric and Polar Functions Lesson 3.6: Sinusoidal Function Transformations (2–3 instructional periods, 55 minutes each)

Primary Objective: Students will identify the amplitude, period, phase shift, and midline of sinusoidal functions of the form $f(\theta) = a \sin(b(\theta + c)) + d$ and $g(\theta) = a \cos(b(\theta + c)) + d$, and connect each parameter to an additive or multiplicative transformation of the parent sine or cosine function. (*Big Idea: Functions*)

Priority Ideas & Skills

AP Skills

1.C *Construct new functions using transformations* — Apply amplitude, period, midline, and phase-shift transformations to build $f(\theta) = a \sin(b(\theta + c)) + d$ from the parent sine function.

2.B *Construct equivalent representations* — Translate between parameter form $a \sin(b(\theta + c)) + d$ and feature descriptions (amplitude, period, midline, phase shift), and rewrite a cosine function as a phase-shifted sine function.

Key Understandings (EKs)

EK 3.6.A.1: $a \sin(b(\theta + c)) + d$ and $a \cos(b(\theta + c)) + d$ (with $a \neq 0$) are sinusoidal transformations of sine/cosine. Because $\cos \theta = \sin(\theta + \pi/2)$, the same transformations apply to both.

EK 3.6.A.2: $\sin(\theta) + d$ shifts the midline to $y = d$ (vertical translation).

EK 3.6.A.3: $\sin(\theta + c)$ shifts the graph $-c$ units horizontally (phase shift).

EK 3.6.A.4: $a \sin(\theta)$ scales amplitude by $|a|$ (vertical dilation).

EK 3.6.A.5: $\sin(b\theta)$ scales period by $\frac{1}{|b|}$; new period = $\frac{2\pi}{|b|}$ (horizontal dilation).

EK 3.6.A.6: $f(\theta) = a \sin(b(\theta + c)) + d$ has amplitude $|a|$, period $\frac{2\pi}{|b|}$, midline $y = d$, phase shift $-c$.

Vocabulary, Concepts, & Theorems

Term	Definition / Note
sinusoidal function	Any function equivalent to $a \sin(b(\theta + c)) + d$ (or cosine form) with $a \neq 0$. A transformation of the parent sine or cosine.
amplitude	$ a $ — half the vertical distance from midline to max (or min). Controls vertical stretch/compression.
midline	The horizontal line $y = d$; the average of max and min output values. Determined by the vertical shift d .
period	Length of one complete cycle; $T = \frac{2\pi}{ b }$. Controlled by the horizontal dilation factor b .
phase shift	Horizontal translation = $-c$ when written in factored form $b(\theta + c)$. Positive c shifts left; negative c shifts right.
vertical dilation	Multiplication of all outputs by a ; changes amplitude and can reflect over the midline when $a < 0$.
horizontal dilation	Replacement of θ by $b\theta$; changes period. $ b > 1$ compresses; $ b < 1$ stretches.

Activate Prior Knowledge & Spiral Review

Spiral topics activated during Warm-Up (first 8–10 minutes):

- **Unit-circle exact values:** Evaluating $\sin \theta$ and $\cos \theta$ at $0, \pi/6, \pi/4, \pi/3, \pi/2$ and their equivalents — fluency needed to verify transformations numerically.
- **Graphs of $y = \sin \theta$ and $y = \cos \theta$:** Students must recall amplitude = 1, period = 2π , midline $y = 0$ from Lessons 3.4–3.5. Today’s lesson stretches and shifts those baseline graphs.
- **Function transformation vocabulary (Units 1–2):** Vertical shifts, vertical stretches, and horizontal shifts of polynomials and exponentials — the same language and algebraic structure apply here.

Hook — Entry Event

Scenario: Project two sinusoidal graphs on the board side-by-side. Graph 1: $y = \sin \theta$ (amplitude 1, period 2π , midline $y = 0$). Graph 2: $y = 3 \sin(2\theta) + 1$ (taller, faster, shifted up). Both look “wavy” — but they are clearly different.

Discussion prompts:

1. “The two graphs are related. What changed? What stayed the same?”
2. “In $y = 3 \sin(2\theta) + 1$, which number makes the wave taller? Which makes it cycle faster? Which shifts it up?”
3. “How would you shift the wave to the right without changing its height or speed?”

Key tension to surface: All sinusoidal functions come from the same parent shape, but four parameters independently control amplitude, period, phase, and midline. Reading these parameters directly from a formula is a powerful modeling tool used throughout science, engineering, and music.

Transition: “Today we’ll dissect the general form $f(\theta) = a \sin(b(\theta + c)) + d$ and learn to identify each of the four features just by looking at the equation.”

Lesson — Instruction Sequence

Step 1 — Vertical Transformations (a and d) (12 min)

Begin with $y = \sin \theta$. Model $y = 3 \sin \theta$: every output is multiplied by 3, so amplitude becomes 3; period and midline unchanged. Then add d : $y = 3 \sin \theta + 2$ shifts the midline to $y = 2$; max is now 5, min is -1 . Formalize EKs 3.6.A.2 and 3.6.A.4. Key questions: what if $a = -2$? (reflection over midline, amplitude still 2) What if $|a| < 1$? (vertical compression toward midline).

Step 3 — The General Form and Summary Table (10 min)

Write $f(\theta) = a \sin(b(\theta + c)) + d$ and label all four parameters. Complete the summary table in guided notes. Work one full example together: $f(\theta) = -2 \sin(\frac{1}{2}(\theta - \pi)) + 3$ — amplitude = 2, period = 4π , phase shift = π (right), midline $y = 3$, reflected ($a < 0$).

Step 2 — Horizontal Transformations (b and c) (12 min)

Model $y = \sin(2\theta)$: replace θ with 2θ ; the cycle completes in π instead of 2π , halving the period. Formalize EK 3.6.A.5: period = $2\pi/|b|$. Introduce c : $y = \sin(\theta + \pi/4)$ shifts the graph *left* by $\pi/4$ (phase shift = $-c = -\pi/4$). Emphasize factoring: $\sin(2\theta + \pi) = \sin(2(\theta + \pi/2))$ — phase shift = $-\pi/2$, not $-\pi$. Students practice factoring two or three examples.

Step 4 — Sine–Cosine Equivalence (8 min)

Establish EK 3.6.A.1: $\cos \theta = \sin(\theta + \pi/2)$. Demonstrate: $2 \cos(3\theta) = 2 \sin(3(\theta + \pi/6))$. Students verify with a table of values at $\theta = 0, \pi/6$. Close with the key idea: *any* sinusoidal function can be written as either a sine or cosine transformation; the choice is often one of convenience.

Pacing note: Steps 1–2 in Period 1; Steps 3–4 and group activity in Period 2; review and exit ticket in Period 3 if needed.

Active Monitoring

- **Phase-shift sign error (most common):** Students read $+c$ as the phase shift rather than $-c$. Prompt: “Rewrite $\sin(2\theta + \pi)$ in factored form $\sin(b(\theta + c))$. What must c be? Then the phase shift is $-c$.”
- **Period vs. b :** Students report b as the period. Ask: “If $b = 4$, the wave repeats 4 times faster than normal. $\text{Period} = 2\pi/4 = \pi/2$, not 4.”
- **Amplitude and sign:** Students report a negative amplitude when $a < 0$. Reinforce: amplitude $= |a|$ is always non-negative; the negative sign causes a reflection.
- **Midline vs. maximum:** Students confuse d with the max output. Prompt: “Midline $= d$ is the average of max and min. $\text{Max} = d + |a|$.”

Group Work & Differentiation

Students work in groups of 3–4 on the tiered activity (teacher-assigned or student self-selected with guidance).

- **Tier R (Reinforce):** Given five equations of the form $a\sin(b\theta) + d$ (no phase shift), students complete a parameter table (amplitude, period, midline, max, min) with sentence frames to guide written explanations.
- **Tier A (Apply):** Given equations in factored form $a\sin(b(\theta + c)) + d$, students identify all four parameters, determine the sine–cosine equivalence, and match each equation to a provided sinusoidal graph card (no sketching).
- **Tier E (Extend):** Given equations in unfactored form (e.g., $-3\cos(2\theta - \pi/3) + 1$), students factor the argument, identify all parameters, write each as a sine function, and write a verbal transformation description in the correct algebraic order.

Individual Work & Assessment

Exit ticket administered independently (final 5–7 minutes of Period 2 or 3).

1. Given $f(\theta) = -3\sin(2(\theta + \frac{\pi}{4})) - 1$, students identify amplitude, period, phase shift, and midline.
2. Multiple-choice: select the equation matching a verbal description of four sinusoidal parameters.

Data use: Sort exit tickets by error type — phase-shift sign errors, period-formula errors, amplitude-sign errors. Use distribution to determine whether a brief re-teach opener is needed at Lesson 3.7.

Reinforcement & Extension

Homework (Lesson 3.6): 6–7 problems covering: identifying all four parameters from sine and cosine equations in factored and unfactored form, computing period and amplitude, identifying phase shift direction, and 2 AP-style multiple-choice items. Optional extension: rewrite a given cosine function as a sine function with an explicit phase shift.

Preview of Lesson 3.7 — Sinusoidal Function Graphs: Ask students: “Now that you can read amplitude, period, midline, and phase shift from an equation, what information would you need to produce the actual graph? In Lesson 3.7 we’ll go in both directions — from equation to graph, and from graph to equation.”